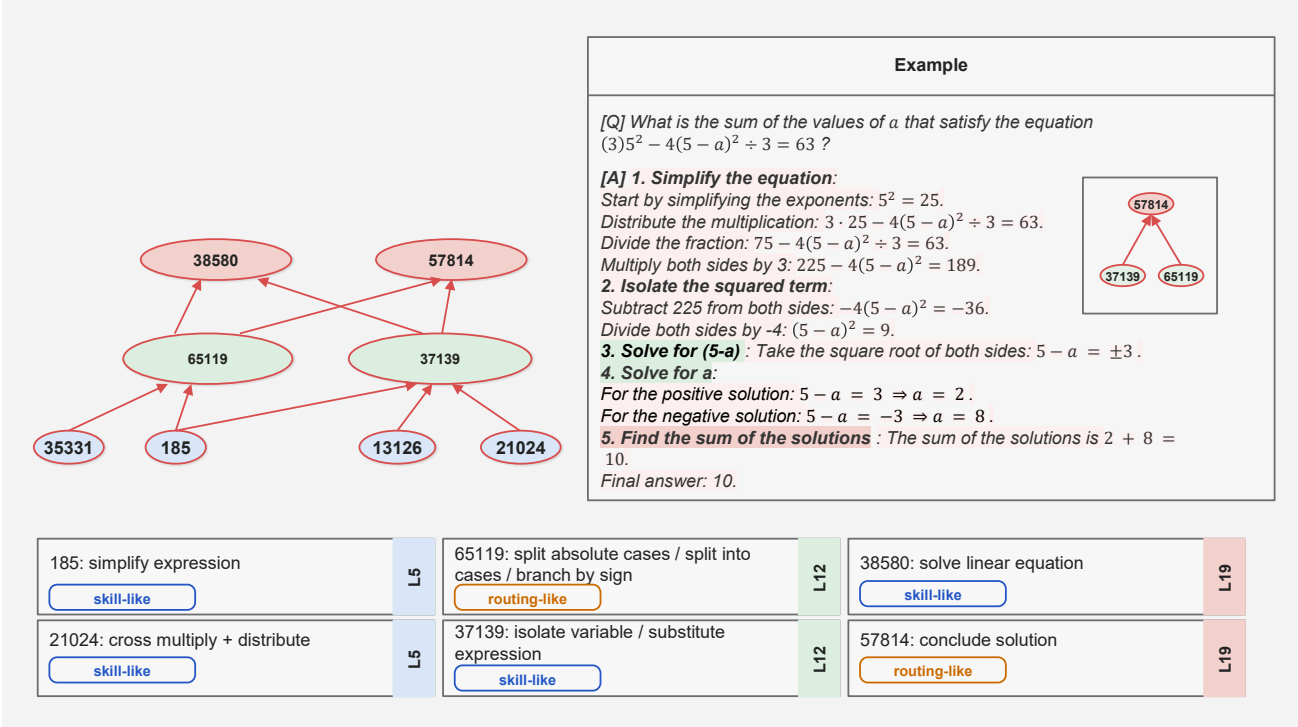

Figure H10. **The learned hierarchical concept graph for autoregressive language models.** Based on real-world math benchmarks, we applied pretrained SAEs to Gemma-2-2b-it, partitioning tokens by sequence position (early, intermediate, final), then ran causal discovery across SAE features. Both the intermediate layer and the high-level layer contain skill-like and routing-like families. The resulting graph reveals a hierarchical structure: a high-level node ("conclude solution") is caused by an intermediate node ("branch by sign"). This shows the hierarchical discrete structure our theory posits does emerge in real transformer representations.

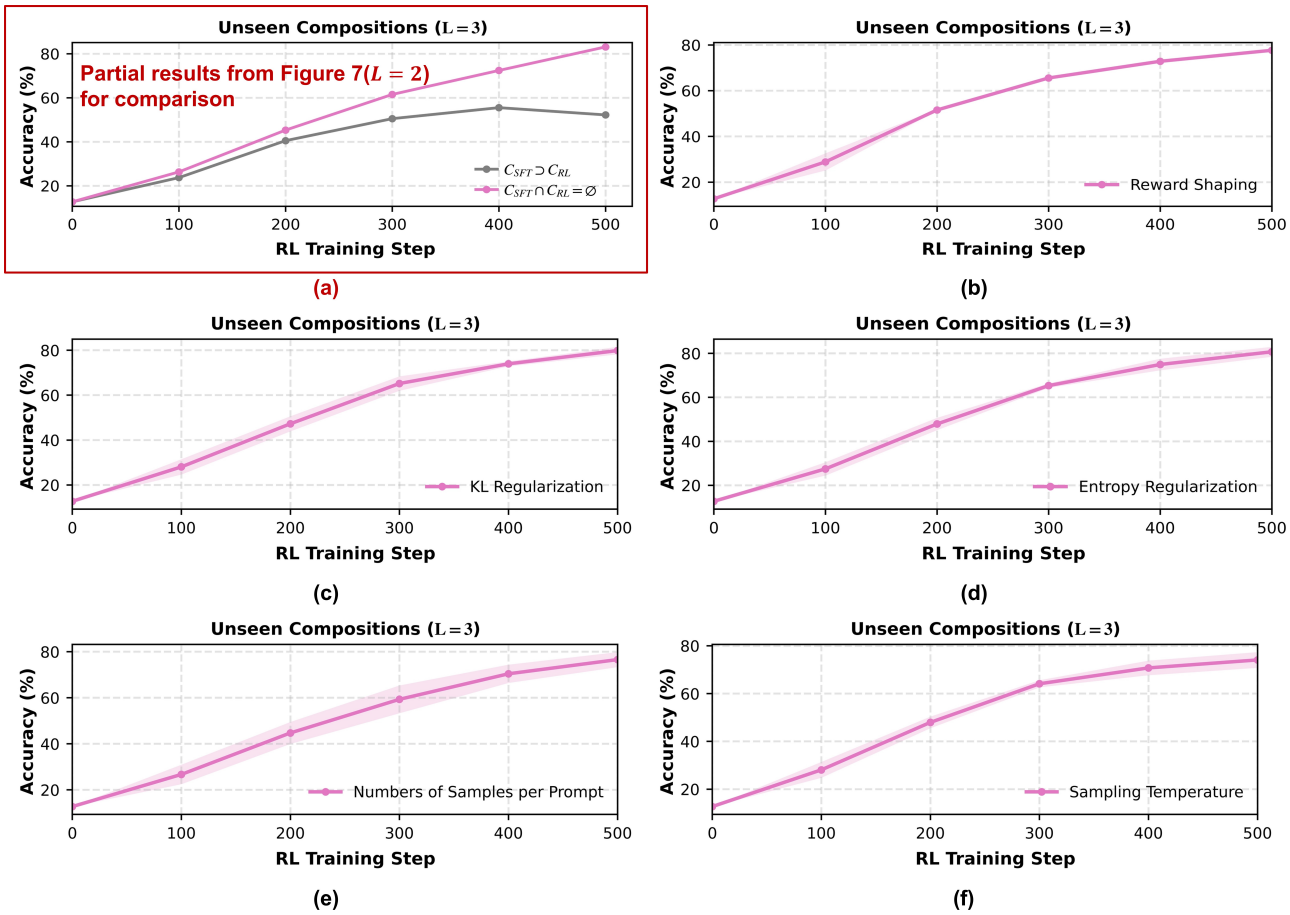

Figure H11. **Comparison of the results of the full sensitivity experiments with partial results from Figure 7 ($L = 2$).** Figure (a) is the partial results from Figure 7 ($L = 2$). Figures (b)-(f) are results of the sensitivity experiments regarding overlong reward shaping (True, False), KL regularization (coefficient=0, 1e-4, 1e-3), entropy regularization (coefficient=0, 5e-4, 1e-3, 2e-3), number of samples per prompt (8, 14, 16), and sampling temperature (0.5, 0.7, 1.0). For each sensitivity experiment, the solid line represents the average result over the same hyperparameter with different values, while the shaded region shows the variation across these runs. The conclusion is that our findings (e.g., Figure 7) are not sensitive to the RL algorithmic details.